

Sample Outputs for taxi-v4 mdp simulation

Environment: taxi-v4

Sample Run 1:

```
Step: 47
Current State: 283
Action Taken: 3
Next State: 263
Reward: -1

Step: 48
Current State: 263
Action Taken: 4
Next State: 263
Reward: -10

Step: 49
Current State: 263
Action Taken: 0
Next State: 363
Reward: -1

Total Reward: -176
```

```
... Step: 42
Current State: 483
Action Taken: 5
Next State: 483
Reward: -10

Step: 43
Current State: 483
Action Taken: 4
Next State: 483
Reward: -10

Step: 44
Current State: 483
Action Taken: 1
Next State: 383
Reward: -1

Step: 45
Current State: 383
Action Taken: 1
Next State: 283
Reward: -1
```

```
Step: 47
Current State: 23
Action Taken: 1
Next State: 23
Reward: -1

Step: 48
Current State: 23
Action Taken: 2
Next State: 23
Reward: -1

Step: 49
Current State: 23
Action Taken: 0
Next State: 123
Reward: -1

Total Reward: -212
```

Sample Run 2:

```
***
Step: 0
Current State: 123
Action Taken: 0
Next State: 223
Reward: -1

Step: 1
Current State: 223
Action Taken: 1
Next State: 123
Reward: -1

Step: 2
Current State: 123
Action Taken: 5
Next State: 123
Reward: -10

Step: 3
Current State: 123
Action Taken: 5
Next State: 123
Reward: -10

Step: 4
Current State: 123
Action Taken: 3
Next State: 103
Reward: -1
```

```
Step: 47  
Current State: 23  
Action Taken: 1  
Next State: 23  
Reward: -1
```

```
Step: 48  
Current State: 23  
Action Taken: 2  
Next State: 23  
Reward: -1
```

```
Step: 49  
Current State: 23  
Action Taken: 0  
Next State: 123  
Reward: -1
```

```
Total Reward: -212
```

Intuition: Taxi-v3 demonstrates sparse and delayed reward behavior in Reinforcement Learning environments. The agent receives small negative rewards for movement (-1), larger penalties for invalid actions (-10), and a positive reward only when the passenger is successfully dropped at the destination (+20).

At this stage, the agent does not possess any intelligence or learning capability because actions are selected randomly using random exploration. Therefore, most sample runs produce poor performance, inefficient movement, and low cumulative rewards.